Micro-Movement and Emotion:

Analysis Methodology & Key Results

```
Dataset:        EmoWear (Kärkkäinen et al.) — 48 participants, lab study
Sensors:        Front STb chest accelerometer (208 Hz) | Empatica E4 wrist (32 Hz)
                Empatica E4 HR/HRV/EDA/SKT | Zephyr BioHarness 3 HR/BR/RR
Observations:   7,104 rows (48 participants × ~37 trials × 4 phases)

Design:  Participants watched 40 short emotion-eliciting video clips, each
         preceded by a baseline rest and followed by a self-report survey
         (valence/arousal/dominance/liking/familiarity, 9-point scales) and
         a ~20 s walking bout.

Question: Do body movement features predict emotional self-reports, and how
          do they compare with physiological measures (HR, HRV, EDA, BR)?

Approach:
  • Feature extraction: per-trial, per-phase segments for all signals
  • Primary movement feature: Animation Index (AI) — high-pass filtered
    accelerometer RMS, adapted from animation/motion capture literature
  • Models: Linear mixed-effects (LME) with random intercepts + slopes
    per participant; predictors z-scored; FDR correction (Benjamini-Hochberg)
  • Sensors analysed separately (STb chest vs E4 wrist)
```

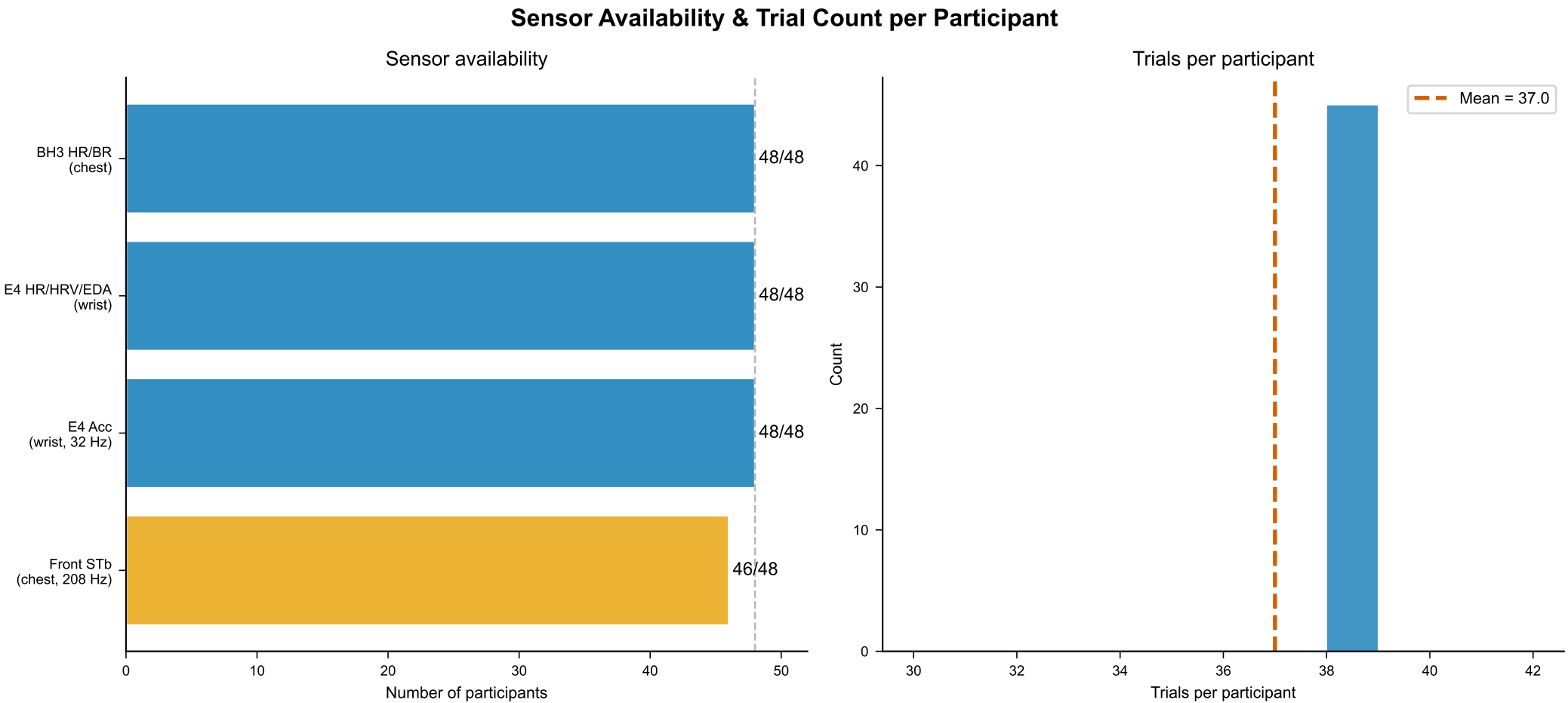
Contents

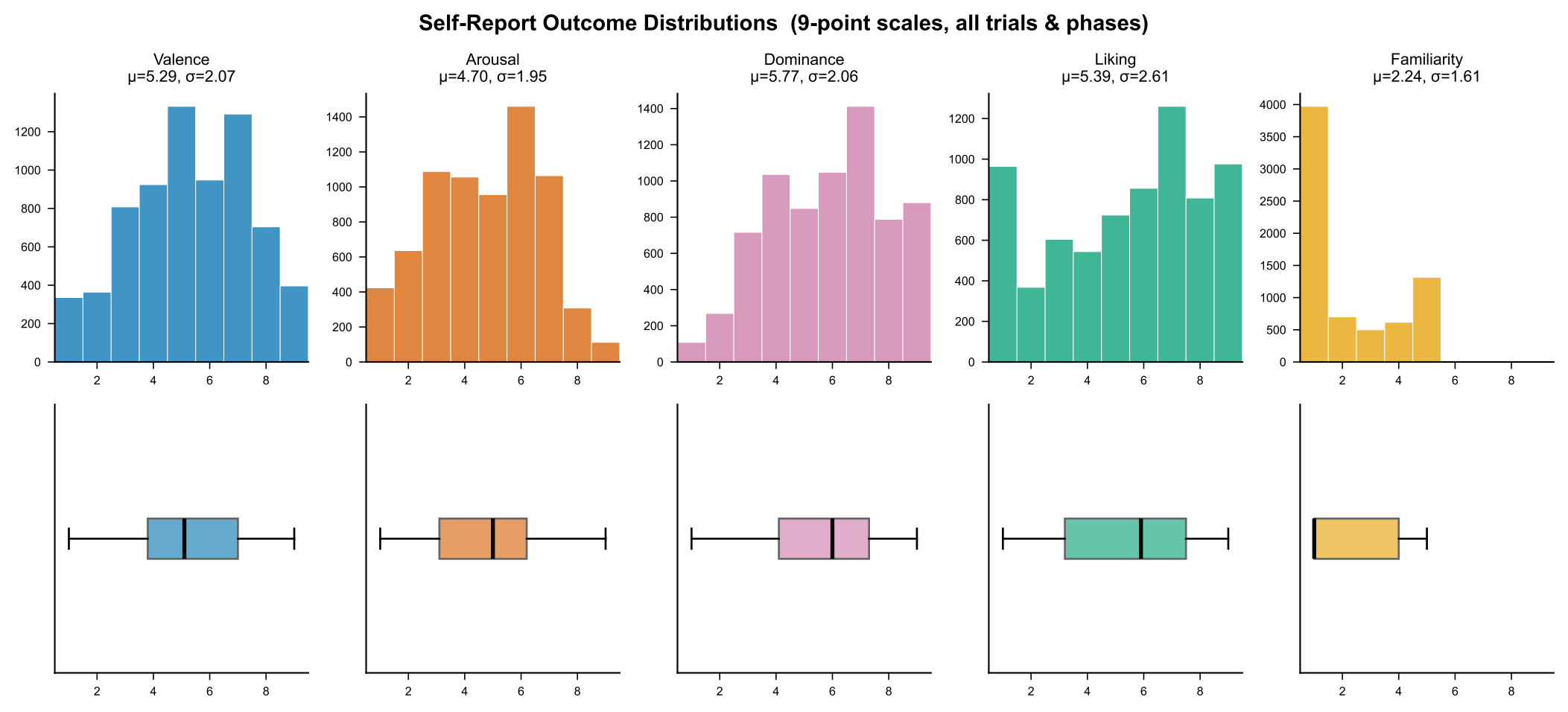
1. Dataset & participant flow
2. Signal preprocessing pipeline
3. Phase segmentation
4. Feature validation (ICC)
5. Statistical framework
6. Main results — forest plot
7. Participant-level evidence
8. Per-participant effect directions
9. Movement vs physiology comparison
10. Model diagnostics

Section 1

Dataset & Participant Flow

What data, from whom, collected how

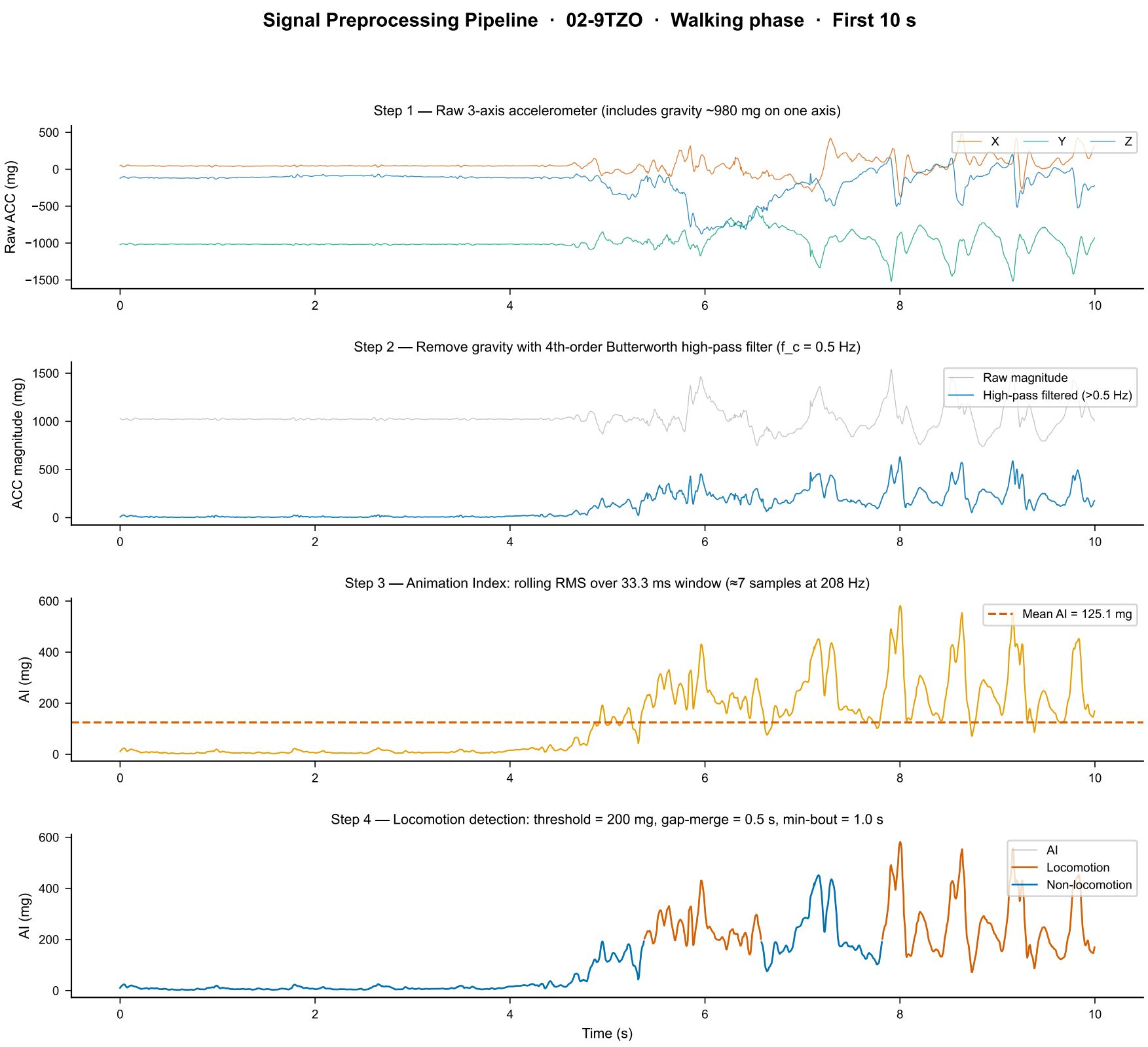

Sensor Availability & Trial Count per Participant
Sensor availability
BH3 HR/BR (chest)
E4 HR/HRV/EDA (wrist)
E4 Acc (wrist, 32 Hz)
Front STb (chest, 208 Hz)
48/48
48/48
48/48
46/48
0
10
20
30
40
50
Number of participants
Trials per participant
Mean = 37.0
Count
0
10
20
30
40
30
32
34
36
38
40
42
Trials per participant


Self-Report Outcome Distributions (9-point scales, all trials & phases)
Valence
μ=5.29, σ=2.07
Arousal
μ=4.70, σ=1.95
Dominance
μ=5.77, σ=2.06
Liking
μ=5.39, σ=2.61
Familiarity
μ=2.24, σ=1.61

Section 2

Signal Preprocessing Pipeline

Raw sensor → time-corrected → phase-segmented → features


Signal Preprocessing Pipeline · 02-9TZO · Walking phase · First 10 s
Step 1 — Raw 3-axis accelerometer (includes gravity ~980 mg on one axis)
Raw ACC (mg)
X
Y
Z
Step 2 — Remove gravity with 4th-order Butterworth high-pass filter (f_c = 0.5 Hz)
ACC magnitude (mg)
Raw magnitude
High-pass filtered (>0.5 Hz)
Step 3 — Animation Index: rolling RMS over 33.3 ms window (≈7 samples at 208 Hz)
AI (mg)
Mean AI = 125.1 mg
Step 4 — Locomotion detection: threshold = 200 mg, gap-merge = 0.5 s, min-bout = 1.0 s
AI (mg)
AI
Locomotion
Non-locomotion
Time (s)

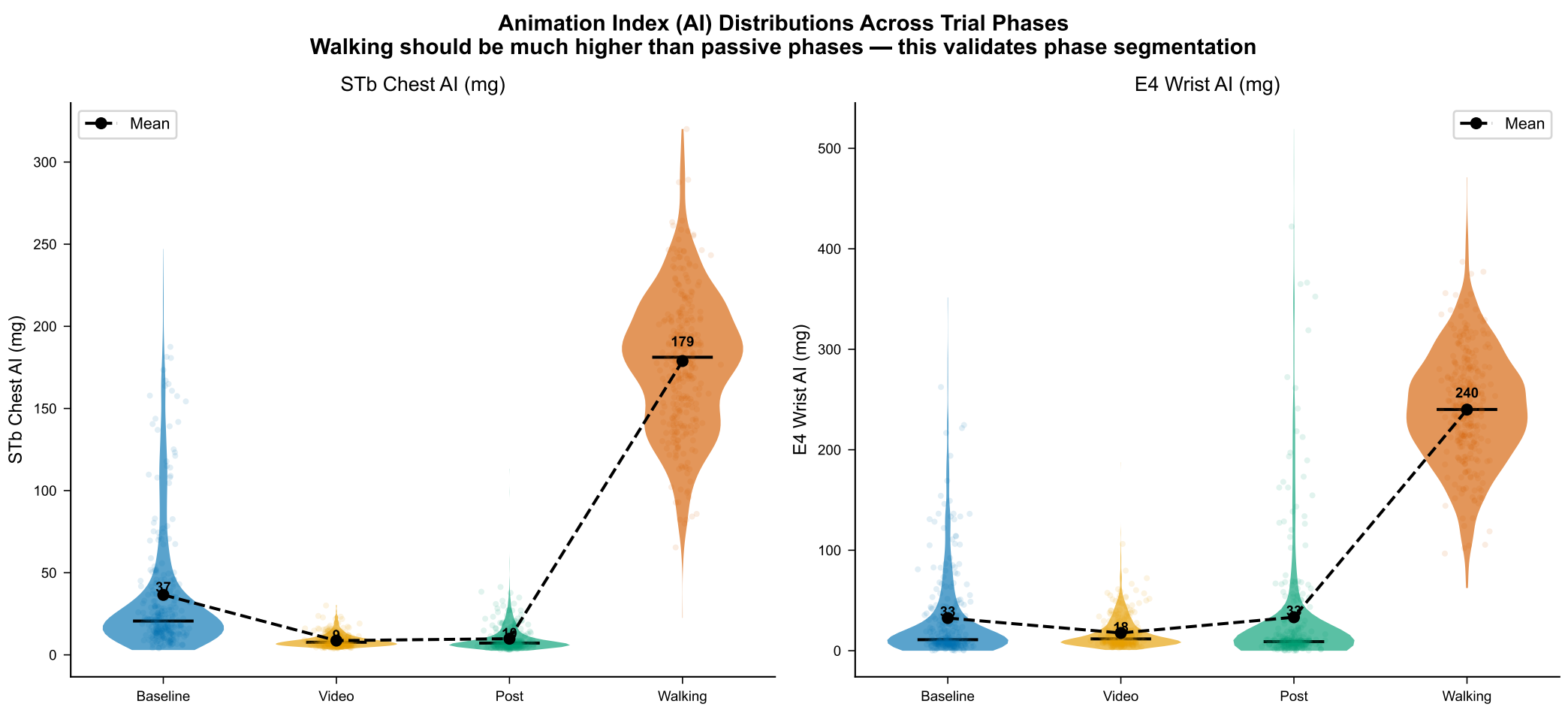
Preprocessing Details

```
TIME CORRECTION

Each device has a clock that drifts from the reference timeline.
Corrected timestamp = raw_timestamp × cf + shift

Device    cf (clock factor)          shift (offset, s)
E4        0.998684  (~1.3 ms/s drift) participant-specific
BH3       1.000000  (no drift)        participant-specific

Parameters loaded per-participant from params.csv.
Without correction, ~5-hour sessions accumulate up to 24 s of E4 drift.

PHASE SEGMENTATION

Phase boundaries are event-marker timestamps from markers-phase2.csv:
  baseline : preB  → vidB      (rest before stimulus)
  video    : vidB  → postB     (stimulus presentation, ~60 s)
  post     : postB → surveyB   (post-stimulus rest)
  walking  : walkB → walkE     (~20 s self-paced walk)

All signal segments are extracted by applying corrected timestamps
to the device-specific signal files.  Only segments ≥ 10 samples are used.

FEATURE EXTRACTION

Per phase-segment, the following are computed:
  STb (208 Hz): ai_mean, ai_std, ai_median, ai_loco_mean, ai_loco_std,
                ai_nonloco_mean, ai_nonloco_std, loco_fraction,
                + 17 movement features (RMS, jerk, spectral, cadence...)
  E4  ( 32 Hz): same feature set on wrist accelerometer
  E4 physio:    HR mean/std, HRV (RMSSD, SDNN, mean IBI), EDA mean/std, SKT
  BH3 physio:   HR mean/std, BR mean/std, HRV from RR intervals
```

Section 3

Feature Distributions by Phase

Validating that the phase segmentation captures real differences


Animation Index (AI) Distributions Across Trial Phases
Walking should be much higher than passive phases — this validates phase segmentation
STb Chest AI (mg)
Mean
STb Chest AI (mg)
0
50
100
150
200
250
300
37
8
10
179
Baseline
Video
Post
Walking
E4 Wrist AI (mg)
Mean
E4 Wrist AI (mg)
0
100
200
300
400
500
33
18
37
240
Baseline
Video
Post
Walking

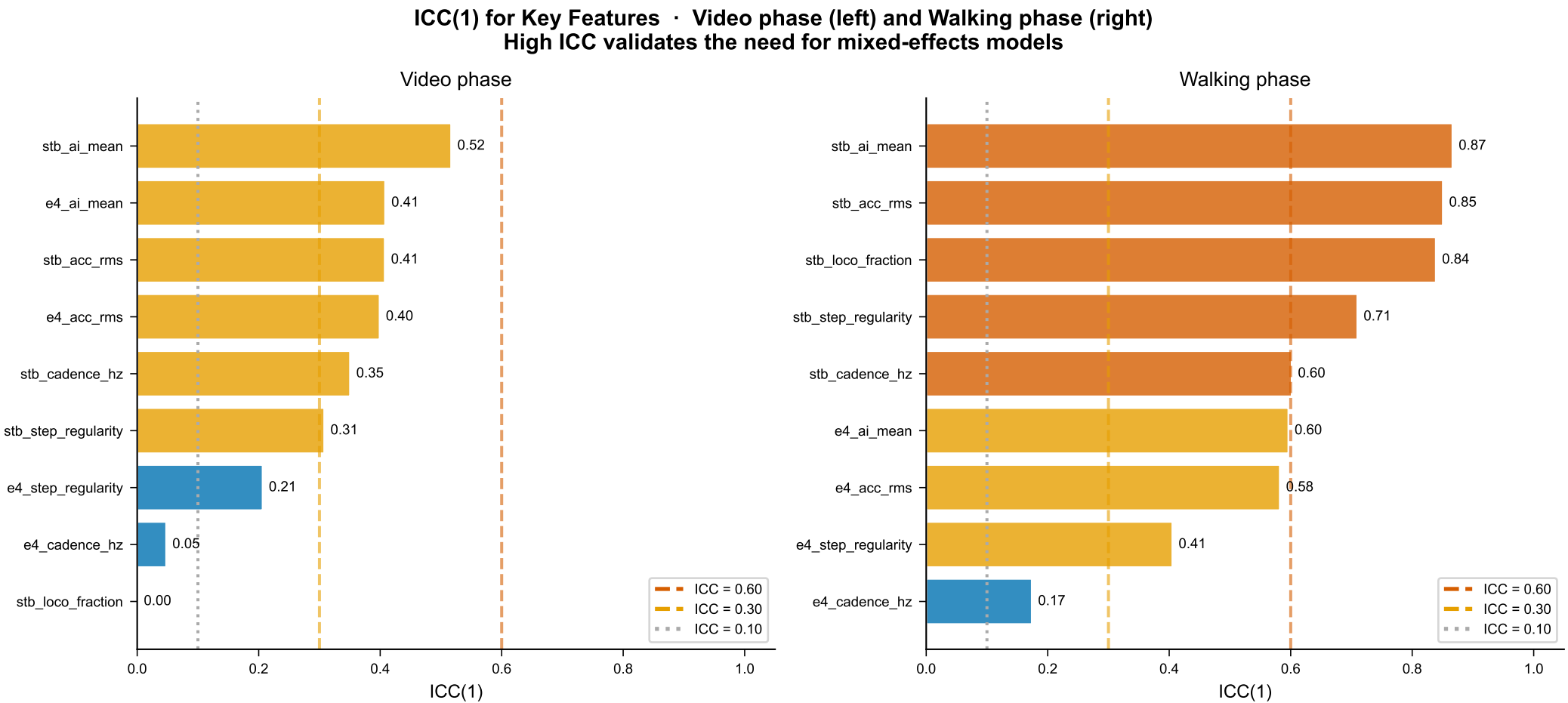
Section 4

Feature Validation — Intraclass Correlations

Justifying mixed-effects models: is there meaningful between-person variance?

Why Mixed-Effects Models?

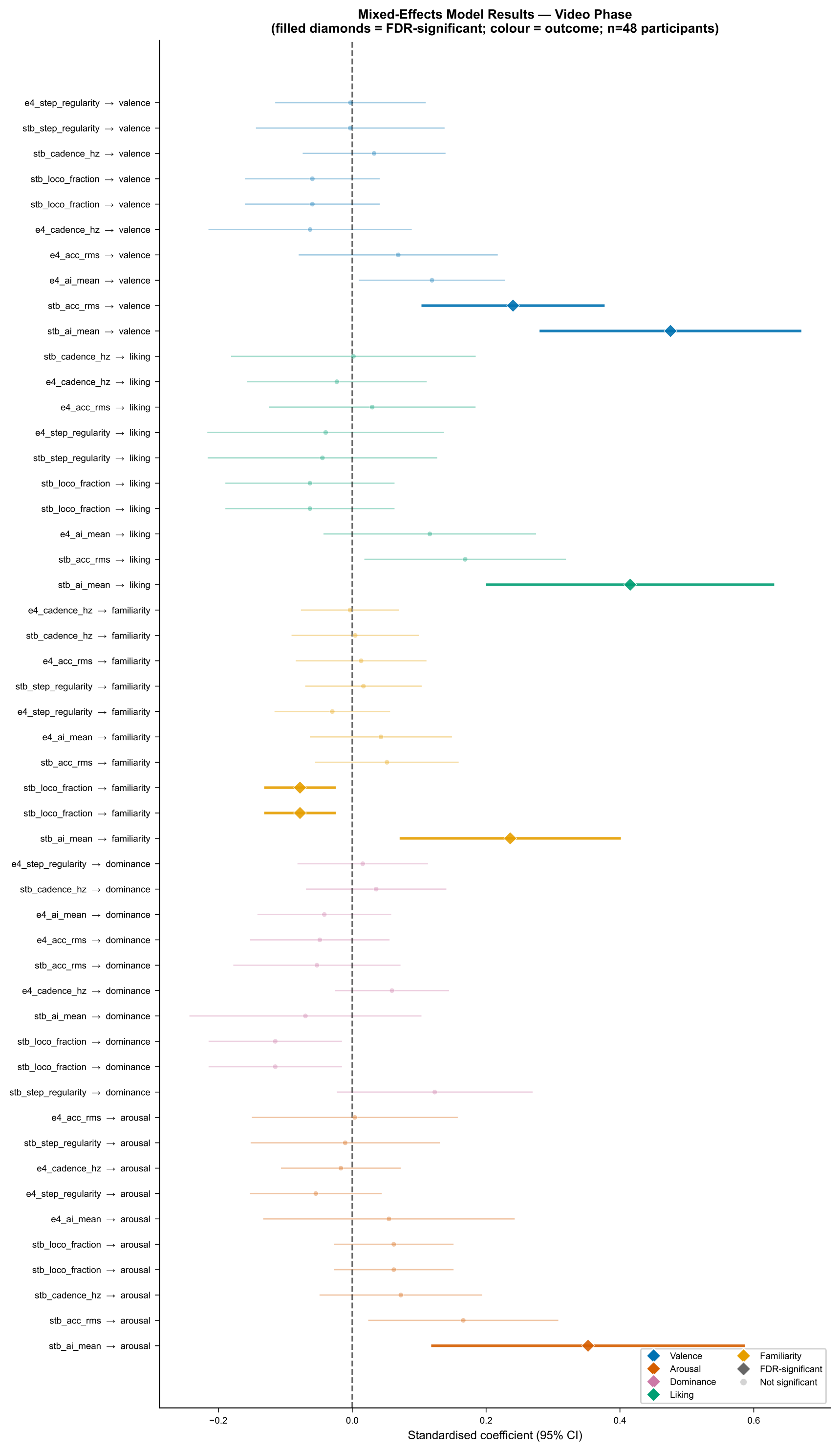
```
INTRACLASS CORRELATION (ICC) — WHY IT MATTERS

A linear mixed-effects model is appropriate when observations are clustered
within participants (i.e., trials from the same person are correlated).

ICC(1) = between-participant variance / total variance

  ICC < 0.10 : negligible clustering → OLS would be adequate
  ICC 0.10–0.30: low clustering     → mixed model preferable
  ICC 0.30–0.60: moderate clustering → mixed model necessary
  ICC > 0.60 : high clustering       → mixed model essential

If ICC is substantial, ignoring clustering inflates Type I error.
Mixed models account for this by allowing each participant their own
intercept (and optionally slope) around the population mean.

MODEL SPECIFICATION

  outcome_z ~ feature_z + (1 + feature_z | participant_id)

  outcome_z and feature_z are z-scored (mean=0, SD=1) within the
  analysis dataset, making coefficients directly comparable across
  outcomes and features (standardised effect sizes).

  Random slopes are included when the model converges; otherwise
  a random-intercept-only model is used (flagged in results as
  model_type = 'random_intercept').

  Optimisation: lbfgs → powell → cg → Nelder-Mead fallback chain.

MULTIPLE COMPARISONS

  Benjamini-Hochberg FDR correction applied across all models
  within each phase (50 tests per phase: 10 features × 5 outcomes).
  Significance threshold: p_fdr < 0.05.
```

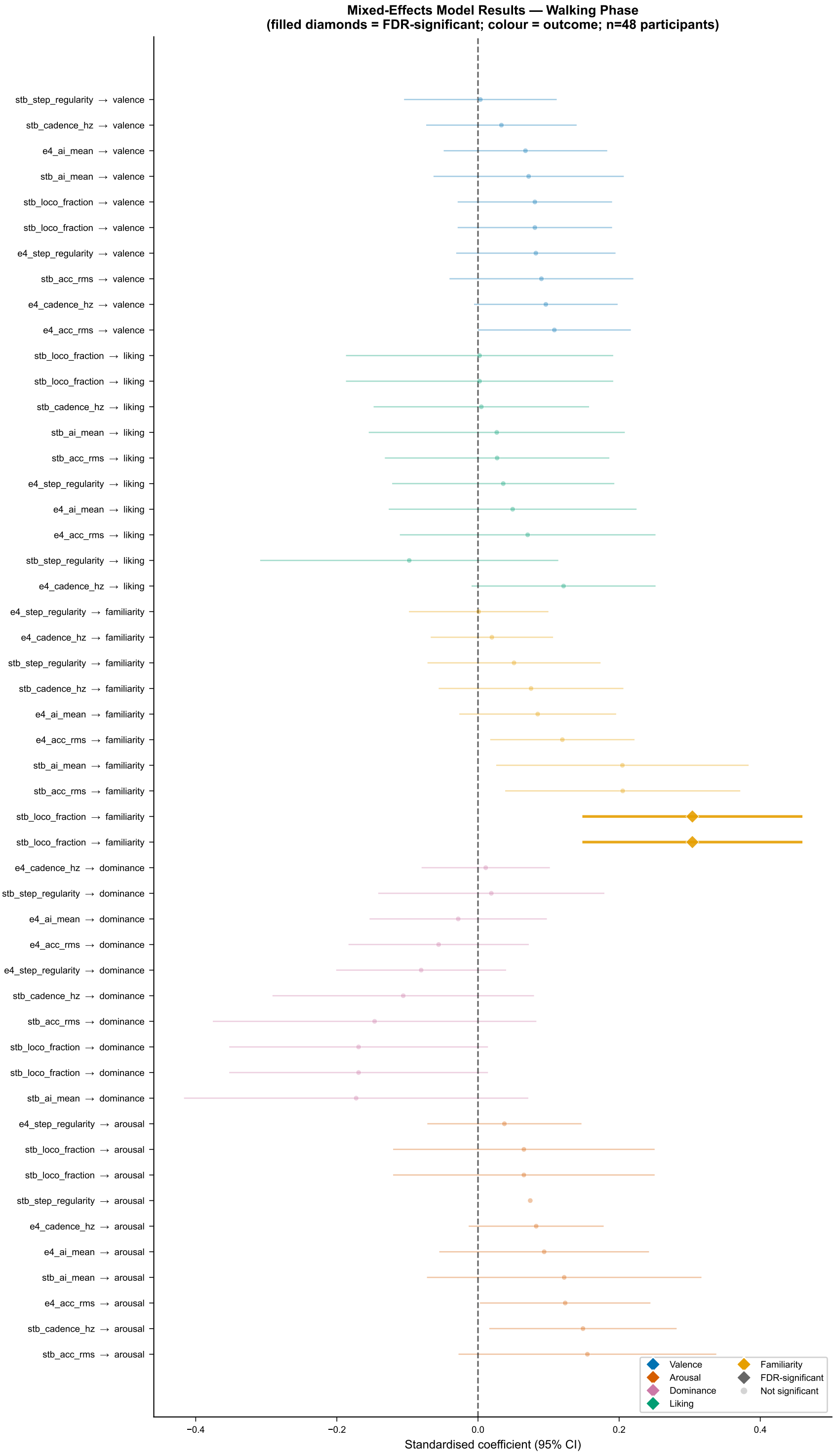

ICC(1) for Key Features · Video phase (left) and Walking phase (right)
High ICC validates the need for mixed-effects models
Video phase
stb_ai_mean
e4_ai_mean
stb_acc_rms
e4_acc_rms
stb_cadence_hz
stb_step_regularity
e4_step_regularity
e4_cadence_hz
stb_loco_fraction
0.52
0.41
0.41
0.40
0.35
0.31
0.21
0.05
0.00
ICC = 0.60
ICC = 0.30
ICC = 0.10
0.0
0.2
0.4
0.6
0.8
1.0
ICC(1)
Walking phase
stb_ai_mean
stb_acc_rms
stb_loco_fraction
stb_step_regularity
stb_cadence_hz
e4_ai_mean
e4_acc_rms
e4_step_regularity
e4_cadence_hz
0.87
0.85
0.84
0.71
0.60
0.60
0.58
0.41
0.17
ICC = 0.60
ICC = 0.30
ICC = 0.10
0.0
0.2
0.4
0.6
0.8
1.0
ICC(1)

Section 5

Main Results — Forest Plot

Standardised coefficients with 95% confidence intervals

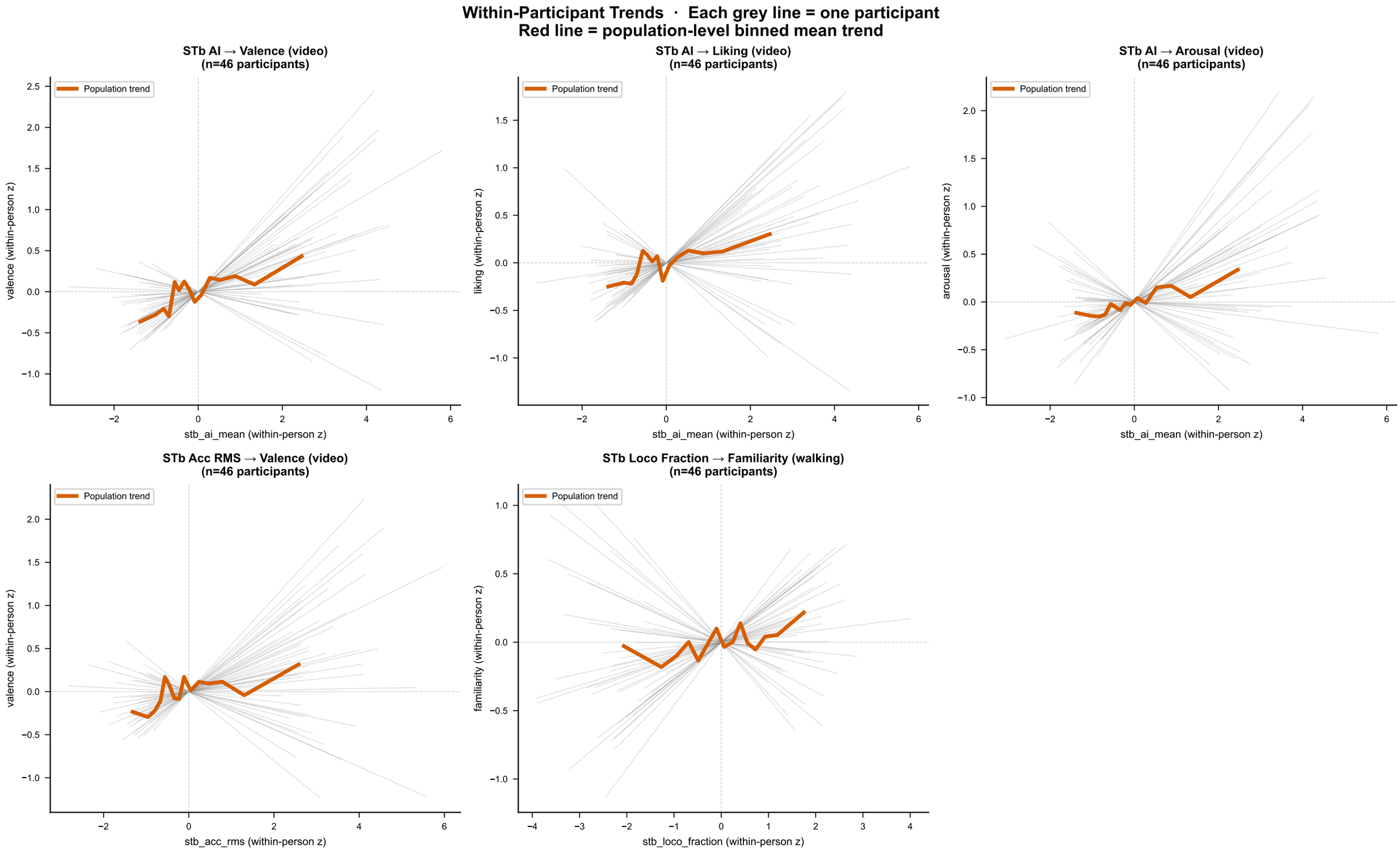

Mixed-Effects Model Results — Video Phase
(filled diamonds = FDR-significant; colour = outcome; n=48 participants)
e4_step_regularity → valence
stb_step_regularity → valence
stb_cadence_hz → valence
stb_loco_fraction → valence
stb_loco_fraction → valence
e4_cadence_hz → valence
e4_acc_rms → valence
e4_ai_mean → valence
stb_acc_rms → valence
stb_ai_mean → valence
stb_cadence_hz → liking
e4_cadence_hz → liking
e4_acc_rms → liking
e4_step_regularity → liking
stb_step_regularity → liking
stb_loco_fraction → liking
stb_loco_fraction → liking
e4_ai_mean → liking
stb_acc_rms → liking
stb_ai_mean → liking
e4_cadence_hz → familiarity
stb_cadence_hz → familiarity
e4_acc_rms → familiarity
stb_step_regularity → familiarity
e4_step_regularity → familiarity
e4_ai_mean → familiarity
stb_acc_rms → familiarity
stb_loco_fraction → familiarity
stb_loco_fraction → familiarity
stb_ai_mean → familiarity
e4_step_regularity → dominance
stb_cadence_hz → dominance
e4_ai_mean → dominance
e4_acc_rms → dominance
stb_acc_rms → dominance
e4_cadence_hz → dominance
stb_ai_mean → dominance
stb_loco_fraction → dominance
stb_loco_fraction → dominance
stb_step_regularity → dominance
e4_acc_rms → arousal
stb_step_regularity → arousal
e4_cadence_hz → arousal
e4_step_regularity → arousal
e4_ai_mean → arousal
stb_loco_fraction → arousal
stb_loco_fraction → arousal
stb_cadence_hz → arousal
stb_acc_rms → arousal
stb_ai_mean → arousal
Valence
Arousal
Dominance
Liking
Familiarity
FDR-significant
Not significant
−0.2
0.0
0.2
0.4
0.6
Standardised coefficient (95% CI)


Mixed-Effects Model Results — Walking Phase
(filled diamonds = FDR-significant; colour = outcome; n=48 participants)
stb_step_regularity → valence
stb_cadence_hz → valence
e4_ai_mean → valence
stb_ai_mean → valence
stb_loco_fraction → valence
stb_loco_fraction → valence
e4_step_regularity → valence
stb_acc_rms → valence
e4_cadence_hz → valence
e4_acc_rms → valence
stb_loco_fraction → liking
stb_loco_fraction → liking
stb_cadence_hz → liking
stb_ai_mean → liking
stb_acc_rms → liking
e4_step_regularity → liking
e4_ai_mean → liking
e4_acc_rms → liking
stb_step_regularity → liking
e4_cadence_hz → liking
e4_step_regularity → familiarity
e4_cadence_hz → familiarity
stb_step_regularity → familiarity
stb_cadence_hz → familiarity
e4_ai_mean → familiarity
e4_acc_rms → familiarity
stb_ai_mean → familiarity
stb_acc_rms → familiarity
stb_loco_fraction → familiarity
stb_loco_fraction → familiarity
e4_cadence_hz → dominance
stb_step_regularity → dominance
e4_ai_mean → dominance
e4_acc_rms → dominance
e4_step_regularity → dominance
stb_cadence_hz → dominance
stb_acc_rms → dominance
stb_loco_fraction → dominance
stb_loco_fraction → dominance
stb_ai_mean → dominance
e4_step_regularity → arousal
stb_loco_fraction → arousal
stb_loco_fraction → arousal
stb_step_regularity → arousal
e4_cadence_hz → arousal
e4_ai_mean → arousal
stb_ai_mean → arousal
e4_acc_rms → arousal
stb_cadence_hz → arousal
stb_acc_rms → arousal
Valence
Arousal
Dominance
Liking
Familiarity
FDR-significant
Not significant
−0.4
−0.2
0.0
0.2
0.4
Standardised coefficient (95% CI)

Section 6

Participant-Level Evidence

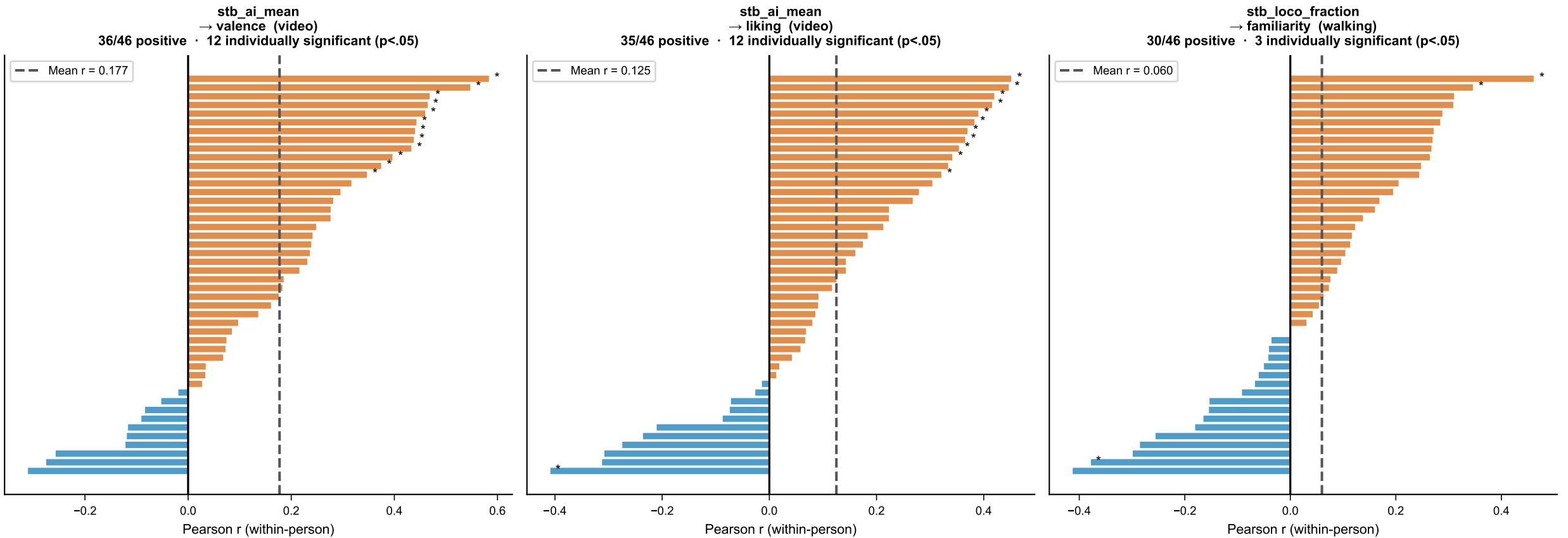
Within-person trends: each line is one participant


Within-Participant Trends · Each grey line = one participant
Red line = population-level binned mean trend
STb AI → Valence (video)
(n=46 participants)
Population trend
valence (within-person z)
stb_ai_mean (within-person z)
STb AI → Liking (video)
(n=46 participants)
Population trend
liking (within-person z)
stb_ai_mean (within-person z)
STb AI → Arousal (video)
(n=46 participants)
Population trend
arousal (within-person z)
stb_ai_mean (within-person z)
STb Acc RMS → Valence (video)
(n=46 participants)
Population trend
valence (within-person z)
stb_acc_rms (within-person z)
STb Loco Fraction → Familiarity (walking)
(n=46 participants)
Population trend
familiarity (within-person z)
stb_loco_fraction (within-person z)

Section 7

Per-Participant Effect Directions

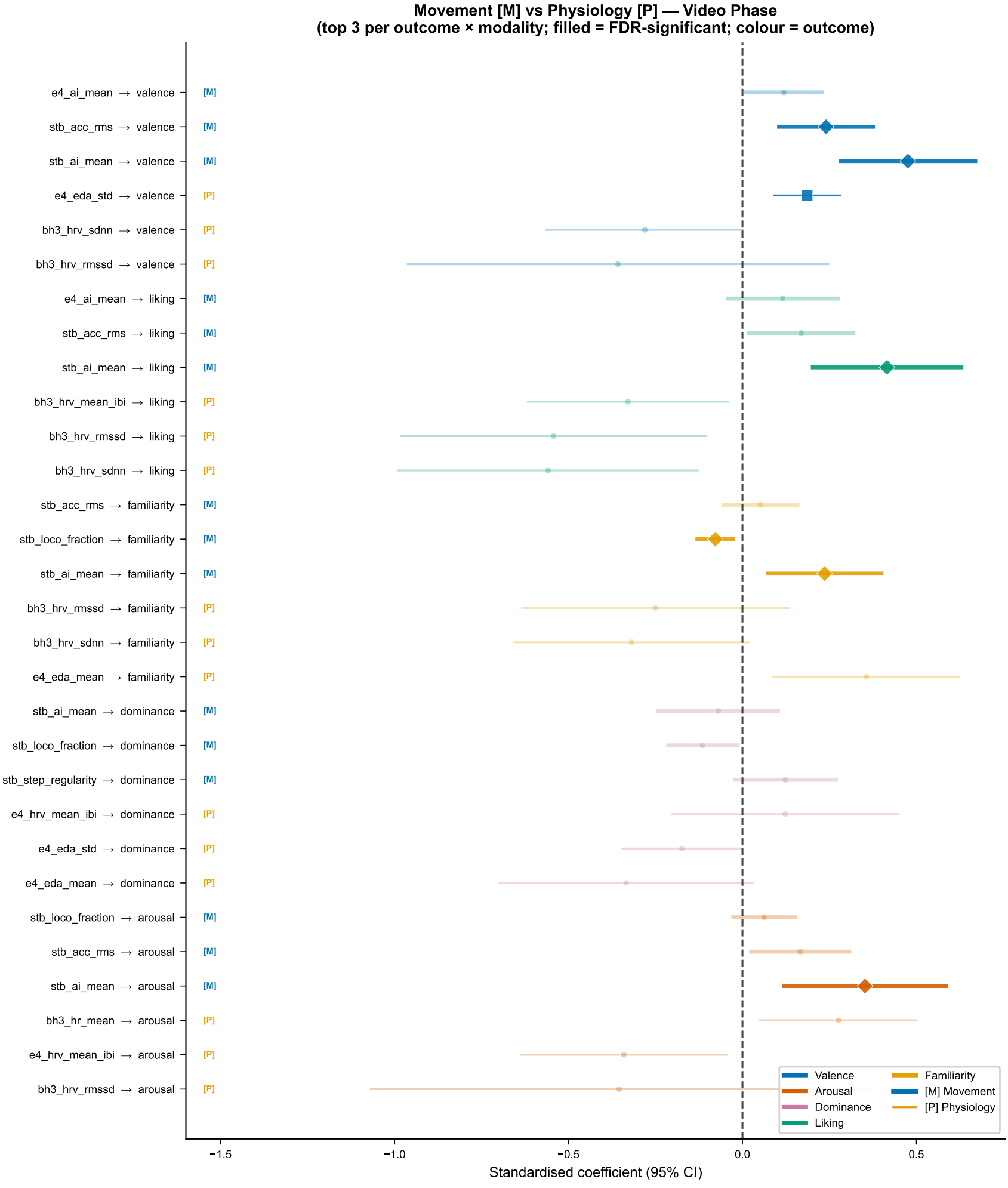
Is the top finding consistent, or driven by a minority of participants?

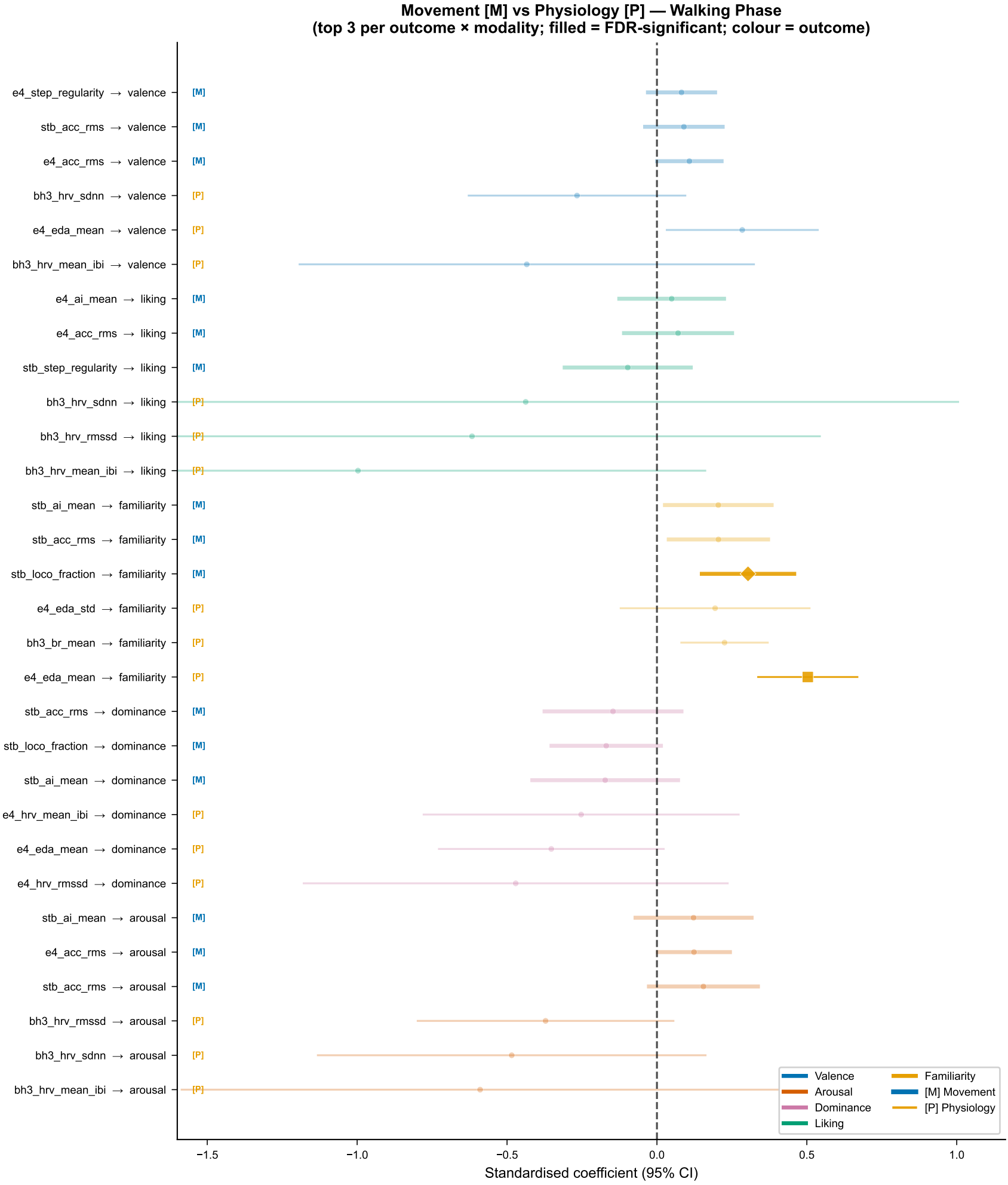

Within-Participant Correlation: Feature → Outcome · Each dot = one participant
Positive = more movement associated with higher rating for that person
stb_ai_mean
→ valence (video)
36/46 positive · 12 individually significant (p<.05)
Mean r = 0.177
Pearson r (within-person)
stb_ai_mean
→ liking (video)
35/46 positive · 12 individually significant (p<.05)
Mean r = 0.125
Pearson r (within-person)
stb_loco_fraction
→ familiarity (walking)
30/46 positive · 3 individually significant (p<.05)
Mean r = 0.060
Pearson r (within-person)

Section 8

Movement vs Physiology — Effect Size Comparison

Standardised coefficients side-by-side across all modalities


Movement [M] vs Physiology [P] — Video Phase
(top 3 per outcome × modality; filled = FDR-significant; colour = outcome)
e4_ai_mean → valence [M]
stb_acc_rms → valence [M]
stb_ai_mean → valence [M]
e4_eda_std → valence [P]
bh3_hrv_sdnn → valence [P]
bh3_hrv_rmssd → valence [P]
e4_ai_mean → liking [M]
stb_acc_rms → liking [M]
stb_ai_mean → liking [M]
bh3_hrv_mean_ibi → liking [P]
bh3_hrv_rmssd → liking [P]
bh3_hrv_sdnn → liking [P]
stb_acc_rms → familiarity [M]
stb_loco_fraction → familiarity [M]
stb_ai_mean → familiarity [M]
bh3_hrv_rmssd → familiarity [P]
bh3_hrv_sdnn → familiarity [P]
e4_eda_mean → familiarity [P]
stb_ai_mean → dominance [M]
stb_loco_fraction → dominance [M]
stb_step_regularity → dominance [M]
e4_hrv_mean_ibi → dominance [P]
e4_eda_std → dominance [P]
e4_eda_mean → dominance [P]
stb_loco_fraction → arousal [M]
stb_acc_rms → arousal [M]
stb_ai_mean → arousal [M]
bh3_hr_mean → arousal [P]
e4_hrv_mean_ibi → arousal [P]
bh3_hrv_rmssd → arousal [P]
Valence
Arousal
Dominance
Liking
Familiarity
[M] Movement
[P] Physiology
−1.5
−1.0
−0.5
0.0
0.5
Standardised coefficient (95% CI)


Movement [M] vs Physiology [P] — Walking Phase
(top 3 per outcome × modality; filled = FDR-significant; colour = outcome)
e4_step_regularity → valence [M]
stb_acc_rms → valence [M]
e4_acc_rms → valence [M]
bh3_hrv_sdnn → valence [P]
e4_eda_mean → valence [P]
bh3_hrv_mean_ibi → valence [P]
e4_ai_mean → liking [M]
e4_acc_rms → liking [M]
stb_step_regularity → liking [M]
bh3_hrv_sdnn → liking [P]
bh3_hrv_rmssd → liking [P]
bh3_hrv_mean_ibi → liking [P]
stb_ai_mean → familiarity [M]
stb_acc_rms → familiarity [M]
stb_loco_fraction → familiarity [M]
e4_eda_std → familiarity [P]
bh3_br_mean → familiarity [P]
e4_eda_mean → familiarity [P]
stb_acc_rms → dominance [M]
stb_loco_fraction → dominance [M]
stb_ai_mean → dominance [M]
e4_hrv_mean_ibi → dominance [P]
e4_eda_mean → dominance [P]
e4_hrv_rmssd → dominance [P]
stb_ai_mean → arousal [M]
e4_acc_rms → arousal [M]
stb_acc_rms → arousal [M]
bh3_hrv_rmssd → arousal [P]
bh3_hrv_sdnn → arousal [P]
bh3_hrv_mean_ibi → arousal [P]
Valence
Arousal
Dominance
Liking
Familiarity
[M] Movement
[P] Physiology
−1.5
−1.0
−0.5
0.0
0.5
1.0
Standardised coefficient (95% CI)

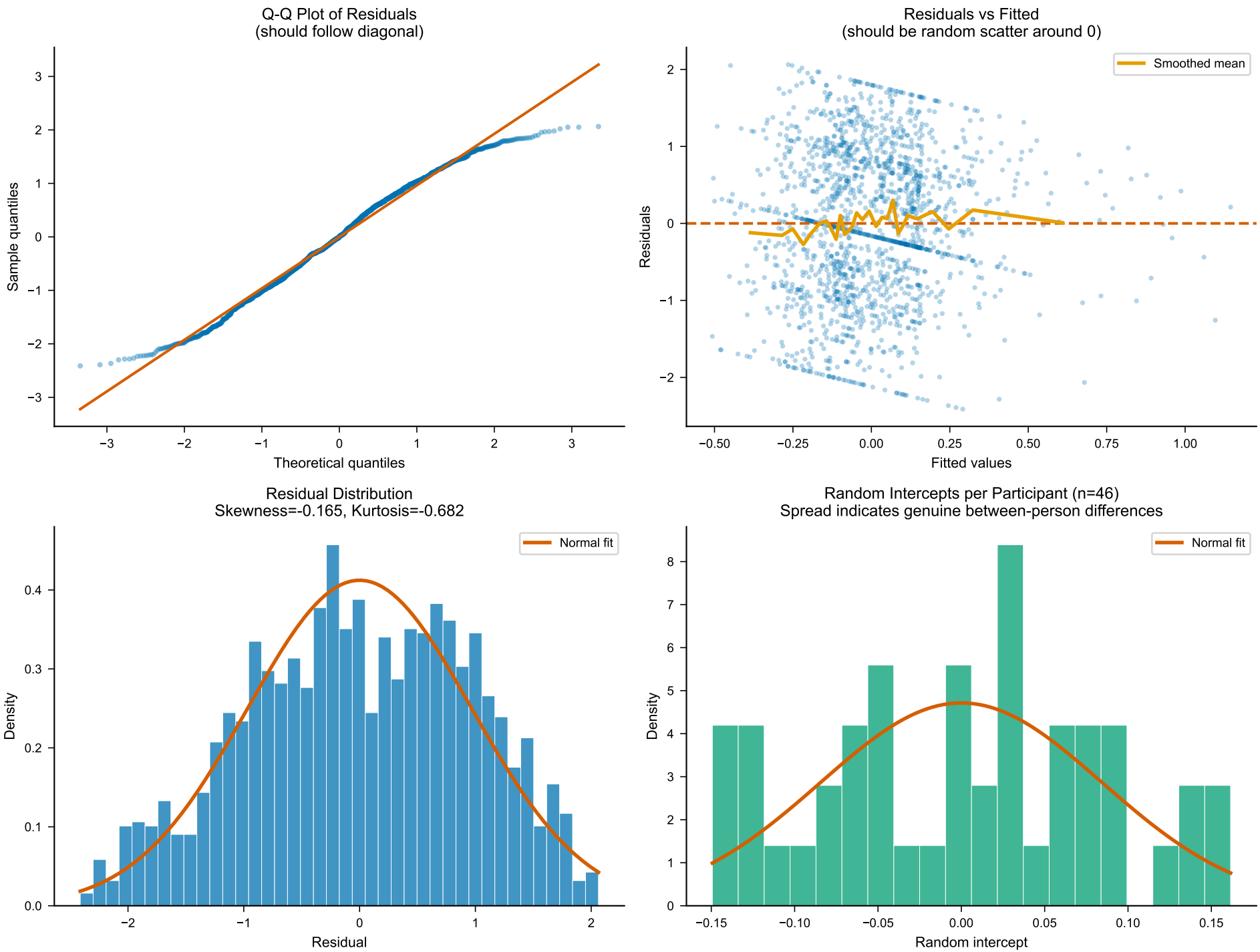
Full Results Summary

```
SIGNIFICANT ASSOCIATIONS AFTER FDR CORRECTION (p_fdr < 0.05)

MOVEMENT:
  [video  ] stb_ai_mean               → valence       β=+0.476  q=0.0001
  [video  ] stb_ai_mean               → liking        β=+0.415  q=0.0024
  [video  ] stb_acc_rms               → valence       β=+0.240  q=0.0056
  [video  ] stb_ai_mean               → arousal       β=+0.352  q=0.0209
  [video  ] stb_loco_fraction         → familiarity   β=-0.078  q=0.0209
  [video  ] stb_ai_mean               → familiarity   β=+0.236  q=0.0269
  [walking] stb_loco_fraction         → familiarity   β=+0.304  q=0.0038

PHYSIOLOGY:
  [video  ] e4_eda_std                → valence       β=+0.186  q=0.0096
  [video  ] bh3_br_mean               → arousal       β=+0.259  q=0.0486
  [walking] e4_eda_mean               → familiarity   β=+0.503  q=0.0000
```

Section 9

Model Diagnostics

Checking LME assumptions for the top model: stb_ai_mean → valence (video)


Model Diagnostics: stb_ai_mean → valence (video phase)
LME assumptions: normality of residuals, homoscedasticity
Q-Q Plot of Residuals
(should follow diagonal)
Sample quantiles
Theoretical quantiles
Residuals vs Fitted
(should be random scatter around 0)
Smoothed mean
Residuals
Fitted values
Residual Distribution
Skewness=-0.165, Kurtosis=-0.682
Normal fit
Density
Residual
Random Intercepts per Participant (n=46)
Spread indicates genuine between-person differences
Normal fit
Density
Random intercept